Basic Matrix Operations
Answer Key

Algebra 2 Systems and Matrices

Quick Answers

1. 2, 4, 7, -3
2. 12, -9, 6, 18
3. 6, 11, 14, 4
4. 3, -5, 8, -5, 8, -8
5. 8, 8, 3, 16
6. 980, 1070
7. 29
8. -4, 5, 20, 17
9. 5, 10
10. $x = 3$, $y = 2$

Curriculum

Level: Algebra 2 Systems and Matrices
HSN-VM.C.6–C.12 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–4 of 5 across 34 tagged checks.

Common wrong answers

6. *If they got 150: added the stock counts to the prices instead of multiplying them*
7. *If they got 41: added the two diagonal products instead of subtracting*

1. Find $A + B$.

Step 1. Both matrices are 2×2, so the sum is defined and is computed entry by corresponding entry.
Step 2. Row 1: $3 + (-1) = 2$ and $-2 + 6 = 4$. Row 2: $5 + 2 = 7$ and $4 + (-7) = -3$.
Step 3. Collect the four entries in their original positions.

$$\boxed{\begin{bmatrix} 2 & 4 \\ 7 & -3 \end{bmatrix}}$$

2. Find $3A$.

Step 1. A scalar multiplies *every* entry — there is no row-or-column selection to make.
Step 2. $3 \cdot 4 = 12$, $3 \cdot (-3) = -9$, $3 \cdot 2 = 6$, $3 \cdot 6 = 18$.
Step 3. The result keeps the same 2×2 shape.

$$\boxed{\begin{bmatrix} 12 & -9 \\ 6 & 18 \end{bmatrix}}$$

3. Find the product AB.

Step 1. A is 2×2 and B is 2×2: the inner dimensions agree, so AB exists and is 2×2.
Step 2. Entry $(1, 1)$ is row 1 of A paired with column 1 of B: $(2)(4) + (1)(-2) = 8 - 2 = 6$. Entry $(1, 2)$: $(2)(3) + (1)(5) = 11$.
Step 3. Row 2 the same way: $(3)(4) + (-1)(-2) = 12 + 2 = 14$, and $(3)(3) + (-1)(5) = 9 - 5 = 4$.

$$\boxed{\begin{bmatrix} 6 & 11 \\ 14 & 4 \end{bmatrix}}$$

4. Find $A - B$.
Step 1. Both matrices are 2×3, so the difference is defined and is taken entry by entry.
Step 2. Row 1: $5 - 2 = 3,\ \ -2 - 3 = -5,\ \ 7 - (-1) = 8.$
Step 3. Row 2: $1 - 6 = -5,\ \ 4 - (-4) = 8,\ \ -3 - 5 = -8.$ Two of these involve subtracting a negative, which adds. $\boxed{\begin{bmatrix} 3 & -5 & 8 \\ -5 & 8 & -8 \end{bmatrix}}$

5. Find $2A + 3B$.
Step 1. Scale each matrix first, then add: scalar multiplication distributes over the entries, so each entry of the answer is $2a_{ij} + 3b_{ij}$.
Step 2. Entry $(1,1)$: $2(1) + 3(2) = 8$. Entry $(1,2)$: $2(-2) + 3(4) = -4 + 12 = 8$.
Step 3. Entry $(2,1)$: $2(3) + 3(-1) = 6 - 3 = 3$. Entry $(2,2)$: $2(5) + 3(2) = 16$. $\boxed{\begin{bmatrix} 8 & 8 \\ 3 & 16 \end{bmatrix}}$

6. Stock value: find SP.
Step 1. S is 2×3 and P is 3×1, so SP is 2×1 — one number per store, which is exactly what a total value should be.
Step 2. North store: $(12)(40) + (8)(25) + (5)(60) = 480 + 200 + 300$.
Step 3. South store: $(9)(40) + (14)(25) + (6)(60) = 360 + 350 + 360$. Each entry is units times price per unit, summed over the three products, so the north store holds 980 dollars of stock and the south store 1070 dollars. $\boxed{\begin{bmatrix} 980 \\ 1070 \end{bmatrix}}$

7. Determinant of M.
Step 1. For a 2×2 matrix the determinant is the main-diagonal product minus the anti-diagonal product.
Step 2. Main diagonal: $(7)(5) = 35$. Anti-diagonal: $(3)(2) = 6$.
Step 3. $35 - 6 = 29$. The determinant is nonzero, so M is invertible. $\boxed{29}$

8. Is AB defined? What size? Compute it.
Step 1. A is 2×3 and B is 3×2. The inner dimensions (3 and 3) match, so AB is defined, and the outer dimensions give its size: 2×2.
Step 2. Each entry pairs a row of A (three numbers) with a column of B (three numbers): $(1)(2) + (2)(-1) + (-1)(4) = 2 - 2 - 4 = -4$, and $(1)(1) + (2)(3) + (-1)(2) = 1 + 6 - 2 = 5$.
Step 3. Row 2: $(3)(2) + (2)(-1) + (4)(4) = 6 - 2 + 16 = 20$, and $(3)(1) + (2)(3) + (4)(2) = 3 + 6 + 8 = 17$.
Manual review: the written size argument is an open response. Accept any wording that matches inner dimensions to conclude the product exists and uses the outer dimensions for its size. $\boxed{\begin{bmatrix} -4 & 5 \\ 20 & 17 \end{bmatrix}}$

9. Find x and y making the matrices equal.
Step 1. Two matrices are equal exactly when every pair of corresponding entries is equal, so this one matrix statement is really four ordinary equations.
Step 2. Entry $(1,1)$: $x + 2 = 7$, so $x = 5$. Entry $(2,2)$: $y - 1 = 9$, so $y = 10$.
Step 3. The other two entries are already identical ($9 = 9$ and $2 = 2$), so nothing else is required. $\boxed{x = 5,\ y = 10}$

10. Write the system as a matrix equation, then solve.

Step 1. The coefficients form A, the variables form the column X, and the constants form the column B:

$$\begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 12 \\ 1 \end{bmatrix}$$

Multiplying the left side back out reproduces $2x + 3y$ and $x - y$, which is the check that the matrix equation says the same thing as the system.

Step 2. From the second row, $x = y + 1$. Substituting into the first: $2(y + 1) + 3y = 12$, so $5y = 10$ and $y = 2$.

Step 3. Then $x = 2 + 1 = 3$. Check in the first equation: $2(3) + 3(2) = 12$ ✓

Manual review: the matrix equation itself is the student's own representation — check that the coefficient matrix rows match the equations in order, including the -1 coefficient of y in the second row. $\boxed{x = 3,\ y = 2}$